27. Special Numbers

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. 1, 4, 9, 16, 25, 36

2. 36

3. 8

4. C — 27

5. 4

6. Yes — 7 × 7 = 49, so 49 is a square number

7. 1, 3, 6, 10

8. C — 10

9. 81

10. C — 27

11. A, C and E (4, 16 and 36 are square numbers; 10 and 20 are not)

12. 11

13. 125

14. A, C and E (3, 10 and 21 are triangular numbers; 7 and 12 are not)

15. B — 13

Standard — Answers

16. 144 ($12^2 = 144$, which is even and between 100 and 150)

17. 49 (the sum of the first n odd numbers is n^2, so the sum of the first 7 odd numbers is $7^2 = 49$)

18. 36

19. A — 28

20. A, C and E (81 is a square number with square root 9, and 81 is odd; B and D are false, since 81 is not a cube number)

21. 6 and 15 (6 + 15 = 21)

22. 13

23. B — 27

24. 36

25. A and D (1 and 36 are both square numbers and triangular numbers; 9, 16 and 49 are square but not triangular)

26. 36, a square number

27. A, C and E (49, 81 and 121 have square roots of 7, 9 and 11, which are all odd and greater than 6; 64 and 100 have even square roots)

28. 28

29. B — 20

30. 121

31. 64

32. 21

33. 8cm

34. B — 25

35. C — 33

Hard — Answers

36. 169 ($13^2 = 169$ is the first square number above 150)

37. 66 (going from the 10th to the 11th term adds 11, so 55 + 11 = 66)

38. A, C and E (6, 10 and 15 are triangular numbers between 5 and 20; 9 and 14 are not)

39. 36 ($36 = 6^2$, and it is also the 8th triangular number: 1+2+3+4+5+6+7+8 = 36)

40. 64 (the sum of the first n odd numbers is n^2, so the sum of the first 8 odd numbers is $8^2 = 64$)

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